

Drowsy Driver Detection

Using Deep Learning

Binary Classification of Driver Alertness from Face Images

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Abstract

- Problem: Drowsy driving is a leading cause of road accidents; automated detection from face imagery can provide a non-intrusive alert mechanism.
- Objective: Compare a custom convolutional neural network trained from scratch against a ResNet-18 network fine-tuned from ImageNet pre-trained weights.
- Dataset: Driver Drowsiness Dataset (DDD) from Kaggle — 41,793 RGB face images, two classes: Drowsy and Non-Drowsy.
- Models: Custom CNN (Conv-BN-ReLU-Pool blocks, FC head) vs ResNet-18 (torchvision, ImageNet pre-training, replaced FC head).
- Evaluation: Accuracy, Precision, Recall, and F1-Score measured on a held-out validation split of DDD.
- Key finding: Both models reached near-perfect accuracy on the same-distribution validation split; limitations of this evaluation are discussed explicitly.
- Contribution: A reproducible PyTorch pipeline comparing scratch training vs transfer learning on a real-world drowsiness dataset.

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01

Motivation

The Road Safety Problem

Why drowsiness detection matters

20 %

of road fatalities linked
to driver fatigue (WHO)

100 k+

drowsy-driving crashes
per year (NHTSA estimate)

3x

higher crash risk when
driving fatigued

- Fatigue impairs reaction time, lane-keeping, and hazard recognition — effects comparable to alcohol impairment.
- Unlike alcohol, fatigue is not easily detected at a roadside stop; an automated in-vehicle system could flag drowsiness before an incident occurs.
- Long-haul truck drivers, night-shift workers, and commuters on sleep-deprived schedules represent the highest-risk populations.
- Camera-based monitoring (face detection, eye closure, head pose) is a low-cost, non-intrusive path to real-time alerting.

Why Automated Detection?

From rule-based methods to learned representations

Traditional Methods

- Require hand-crafted features: eye aspect ratio (EAR), PERCLOS, head pose angles.
- Sensitive to lighting changes, glasses, and camera angle.
- Feature engineering demands domain expertise and repeated tuning per environment.
- Difficult to generalise across diverse driver populations.

Deep Learning Approach

- Learns discriminative features directly from pixel data during training.
- Convolutional layers capture spatial patterns: eye state, head tilt, expression.
- Transfer learning from large datasets (ImageNet) enables strong initialisation with modest labelled data.
- End-to-end pipeline: raw image in, class label out — no manual feature selection.

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Research Questions

Research Questions

RQ 1

Can a custom CNN trained entirely from scratch on DDD classify driver drowsiness with high accuracy on a same-distribution validation split?

RQ 2

Does fine-tuning a ResNet-18 network pre-trained on ImageNet outperform the custom CNN in terms of accuracy, precision, recall, and F1-score?

RQ 3

How many training epochs does each model require to reach convergence, and what does this suggest about practical training cost?

RQ 4

What are the evaluation limitations of reporting metrics on a validation split drawn from the same dataset, and how should results be interpreted cautiously?

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The Dataset

Dataset Overview

Driver Drowsiness Dataset (DDD) · Kaggle

41,793

Total RGB images

22,348

Drowsy class

19,445

Non-Drowsy class

1.149 : 1

Class ratio

227 × 227

Image size (px)

DDD — one sample per class

Drowsy



Non Drowsy



Figure 1: Sample images — Drowsy (left) and Non-Drowsy (right) from DDD

Class Distribution

Mild imbalance in favour of the Drowsy class

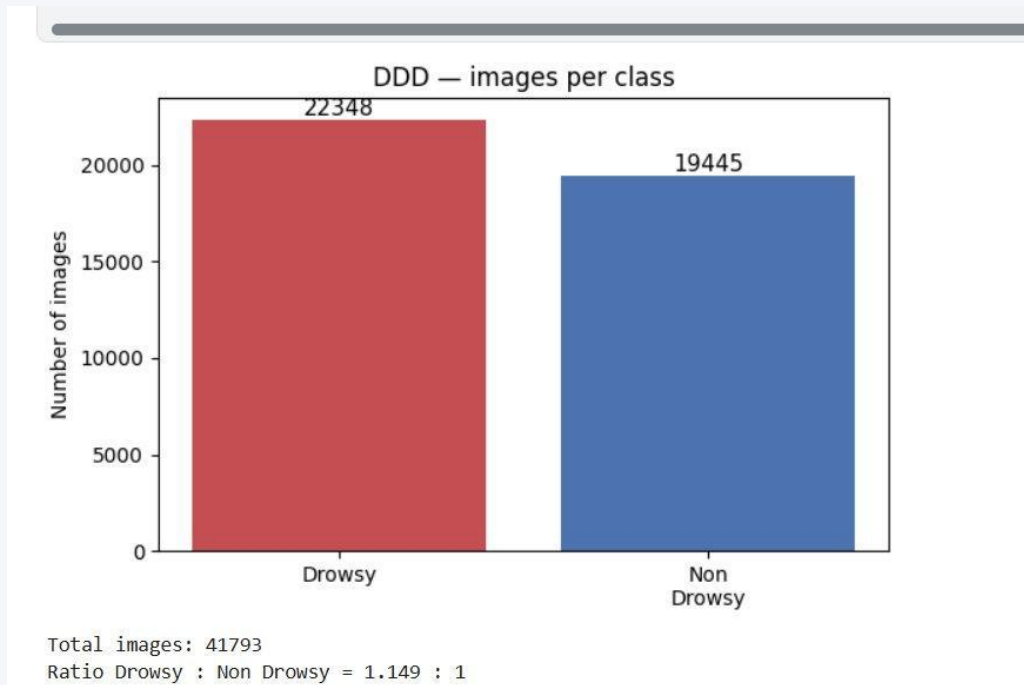


Figure 2: DDD class distribution — Drowsy: 22,348 / Non-Drowsy: 19,445 / Ratio 1.149 : 1

- The Drowsy class is approximately 15 % larger than Non-Drowsy. Both models are trained without explicit oversampling; each sample is weighted equally in the cross-entropy loss.
- The mild imbalance is unlikely to dominate model behaviour at this ratio, but could slightly bias decision thresholds.

Data Preprocessing

Standardisation applied before training both models

1

Resize

All images resized from 227×227 px to 224×224 px. This is the standard input size for torchvision models and reduces memory overhead.

2

Augmentation (train only)

RandomHorizontalFlip, RandomRotation ($\pm 10^\circ$), and ColorJitter applied during training to increase variation and reduce overfitting. Not applied to validation set.

3

Normalisation

All splits normalised with ImageNet statistics: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$. Required for ResNet-18 pre-trained compatibility.

4

DataLoader

PyTorch DataLoader used with batch size 32, shuffle enabled for training split, disabled for validation split.

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Methodology Pipeline

Methodology Overview

End-to-end pipeline from raw data to evaluation



- Both models share the same dataset, split, and preprocessing pipeline — the only variable is the model architecture.
- The validation split is drawn randomly from DDD (stratified). No external test corpus is available; this is addressed in the limitations discussion.

Train–Validation Split



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Model A — Custom CNN

Convolution — Core Intuition

How a convolutional layer detects local patterns

- A convolutional kernel (filter) is a small matrix — e.g. 3×3 — that slides across the input image.
- At each position the kernel performs element-wise multiplication and sums the result, producing one output value.
- This operation detects a specific local pattern (edge, corner, colour gradient) at every position in the image.
- Multiple kernels are stacked in one convolutional layer, each learning to detect a different pattern — producing a volume of feature maps.
- Deeper layers combine low-level detectors into higher-level concepts: edges $\rightarrow$ shapes $\rightarrow$ facial features.
- BatchNorm stabilises training; ReLU introduces non-linearity; MaxPool reduces spatial size, retaining dominant signals.

Input patch

12	20	5
8	45	3
15	7	30

Kernel

1	0	-1
1	0	-1
1	0	-1

$\otimes$ =

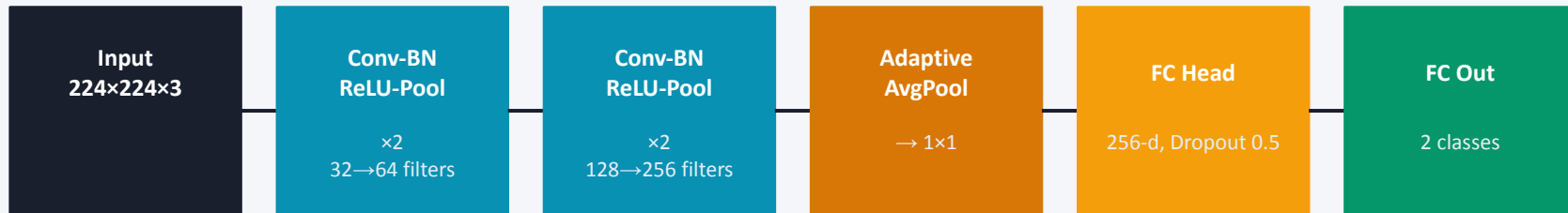
Max Pooling — purpose



Takes maximum value in each region. Reduces spatial size, retains strongest activations.

Custom CNN Architecture

Trained from scratch on DDD



- Four convolutional blocks, each composed of Conv2d, BatchNorm2d, ReLU, and MaxPool2d, extract hierarchical spatial features progressively.
- Filter counts grow: 32 → 64 (blocks 1–2) then 128 → 256 (blocks 3–4), increasing representational capacity at deeper layers.
- Adaptive average pooling collapses the spatial dimensions to 1 × 1, making the architecture independent of exact input size.
- A fully-connected head maps the 256-dimensional representation to 2 class logits; Dropout ($p = 0.5$) provides regularisation.
- Trained with cross-entropy loss, Adam optimiser ($\text{lr} = 1 \times 10^{-3}$), cosine annealing schedule, for 12 epochs.

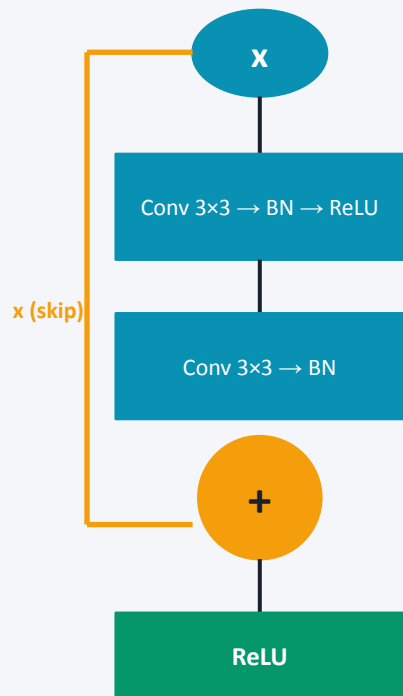
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Model B — ResNet-18

Residual Blocks & Skip Connections

He et al. CVPR 2016 — the core innovation of ResNet

- Gradient vanishing: in very deep networks, gradients become negligible as they propagate backward through many layers.
- Solution — skip connections carry the original input x forward, bypassing one or more layers.
- The layer is asked to learn the residual $F(x) = \text{output} - x$, which is easier to optimise than learning the full mapping from scratch.
- Identity formula: $y = F(x, \{W_i\}) + x$, where F is the residual function and no extra parameters are added.
- ResNet-18 stacks 8 residual blocks (2 layers each) for 18 weight layers total.



ResNet-18 — Transfer Learning Adaptation

Torchvision pre-trained weights · Fine-tuned on DDD

- ResNet-18 is loaded from torchvision with ImageNet pre-trained weights (top-1 accuracy $\approx 69.8\%$). These weights encode low-level texture and edge detectors applicable to any image recognition task.
- The original final fully-connected layer ($512 \rightarrow 1000$ ImageNet classes) is removed and replaced with a $512 \rightarrow 2$ linear layer matching our binary classification head.
- All layers are unfrozen — full end-to-end fine-tuning allows the entire network to adapt its representations to drowsiness-relevant facial cues.
- Because the starting weights are already well-optimised, the network converges within 8 epochs, compared to 12 required by the CNN trained from scratch.
- Optimiser: SGD with momentum 0.9, initial learning rate 1×10^{-3} , cosine annealing schedule. The lower end-of-training LR helps preserve fine-grained ImageNet features.

ResNet-18

Layers:	18
Res Blocks:	8
Params:	~11 M
Pretrain:	ImageNet
Head:	$512 \rightarrow 2$
Epochs:	8
Optim:	SGD
LR start:	$1e-3$

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Training Setup

Training Objective & Learning Rate

Cross-entropy loss · Cosine annealing schedule

Cross-Entropy Loss (C = 2 classes)

$$L = - \sum_{i=1}^N \sum_{c=1}^2 y_{ic} \cdot \log p(y_{ic} | x_i; \theta) \quad \text{where } c \in \{ \text{Drowsy, Non-Drowsy} \}$$

Cosine Annealing LR (PyTorch CosineAnnealingLR)

$$\eta_t = \eta_{\min} + \frac{1}{2} (\eta_{\max} - \eta_{\min}) \cdot (1 + \cos(\pi t / T))$$

Hyperparameter	Custom CNN	ResNet-18
Optimiser	Adam	SGD (mom. 0.9)
Initial LR	1×10^{-3}	1×10^{-3}
LR Schedule	Cosine Annealing	Cosine Annealing
Batch Size	32	32
Epochs	12	8

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Results

Results — CNN Training Curves

Validation loss and accuracy over 12 epochs

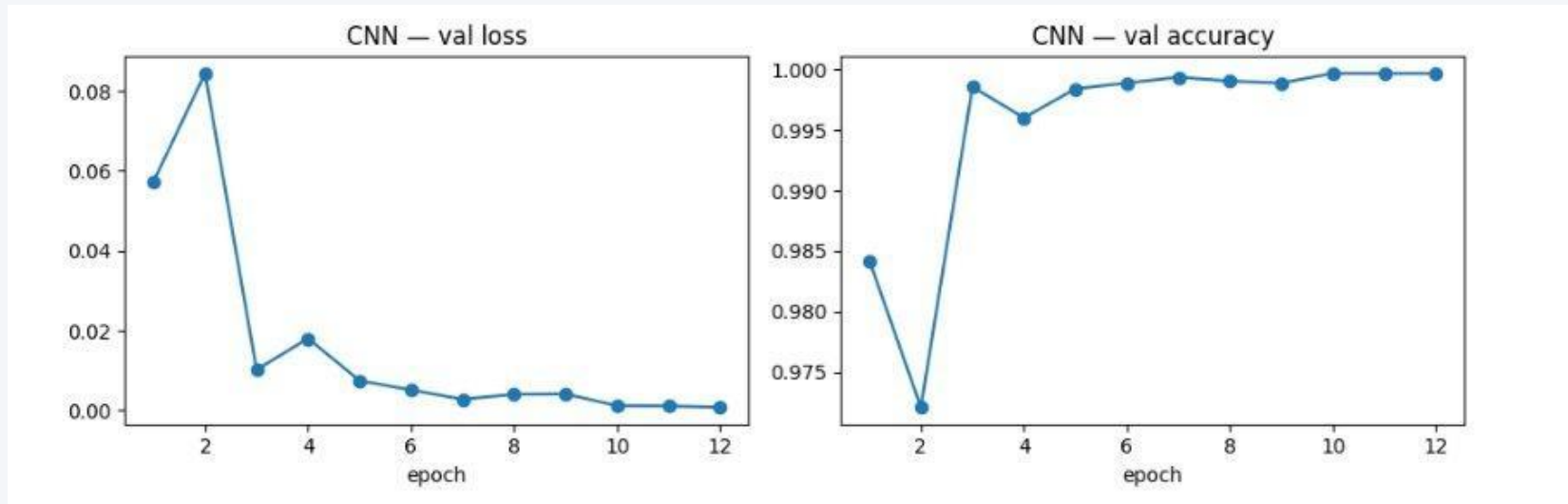


Figure 3: CNN — validation loss (left) and validation accuracy (right) over 12 epochs

- Validation loss peaks transiently at epoch 3, then decreases sharply and stabilises near zero from epoch 8 onward.
- Validation accuracy recovers from a dip at epoch 3 to approximately 99.9 % from epoch 5 onward.
- Validation loss does not increase at any later epoch — no measurable overfitting detected within 12 epochs.

Results — ResNet-18 Training Curves

Validation loss and accuracy over 8 epochs

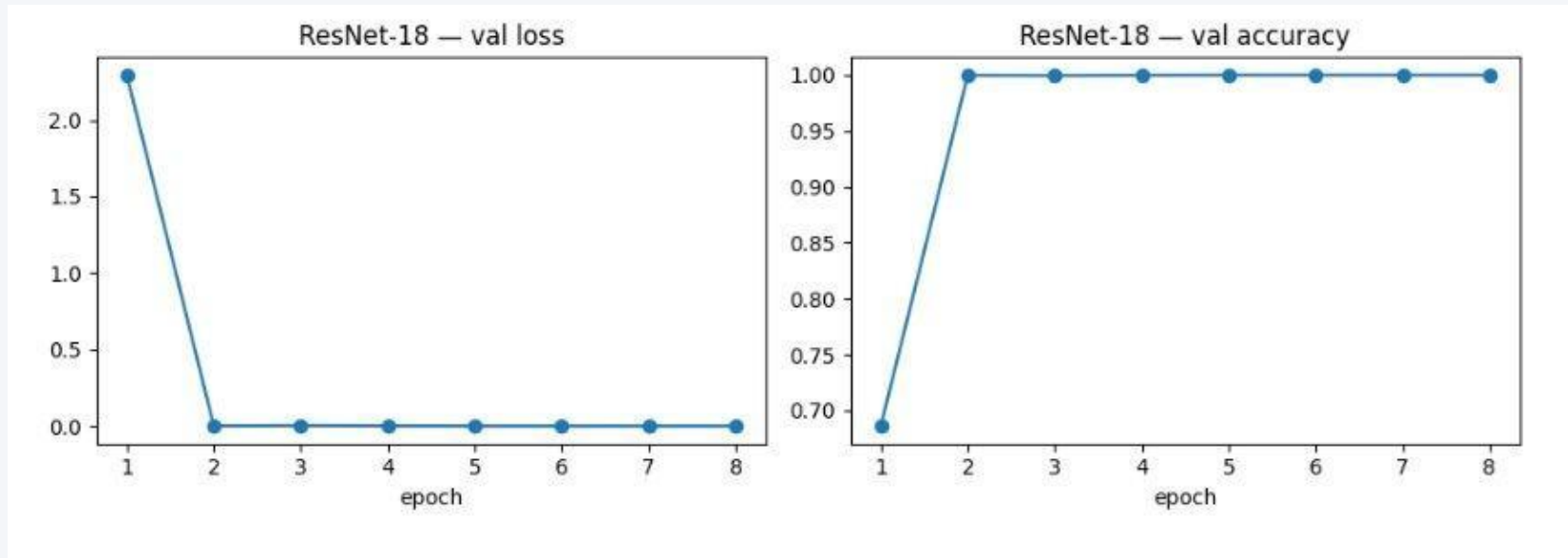


Figure 4: ResNet-18 — validation loss (left) and validation accuracy (right) over 8 epochs

- Validation loss drops from approximately 2.25 at epoch 1 to near zero by epoch 2, then remains flat — rapid convergence attributable to ImageNet pre-training.
- Validation accuracy rises from ~69 % at epoch 1 to ~99.9 % at epoch 2 and holds that level through epoch 8.
- Convergence within a single epoch demonstrates the practical advantage of initialising from pre-trained weights.

Results — Confusion Matrices

CNN (left) vs ResNet-18 (right) · Validation Set

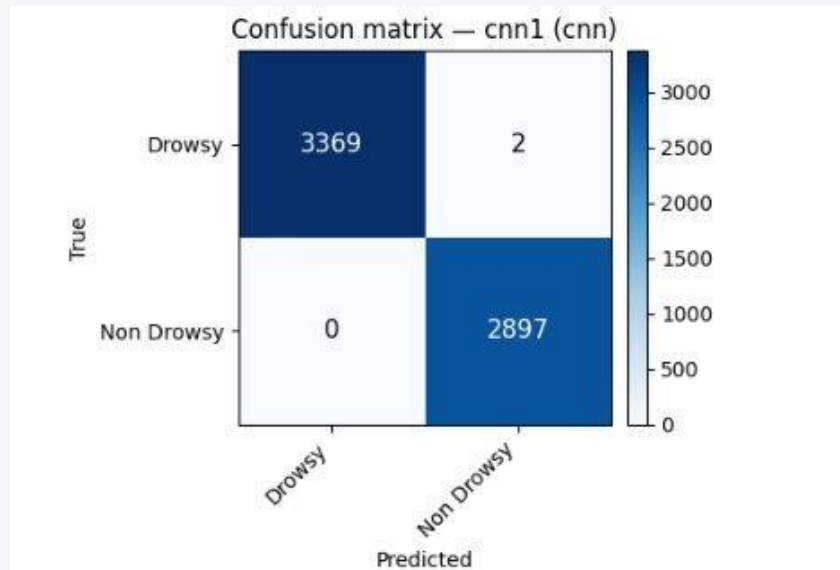


Figure 5: CNN confusion matrix

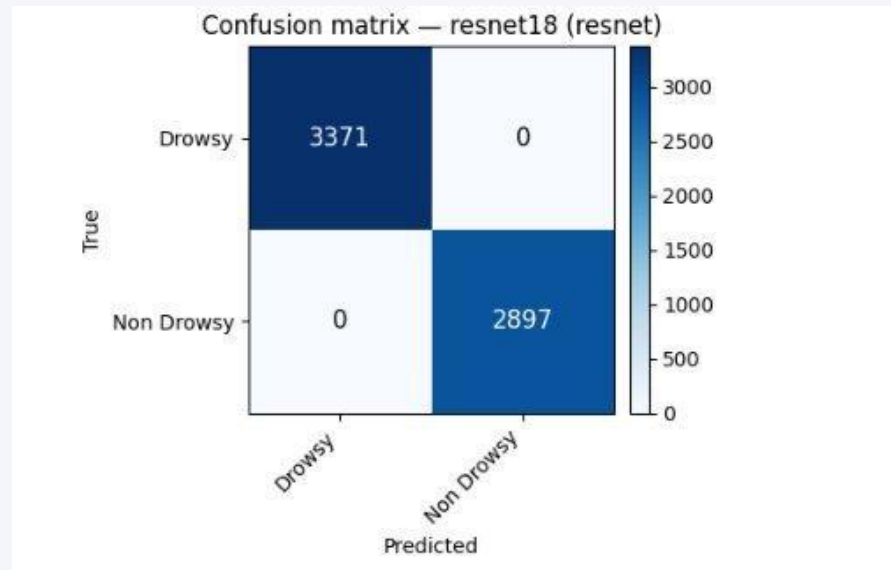


Figure 6: ResNet-18 confusion matrix

CNN: 3,369 TP · 2,897 TN · 2 FN · 0 FP — two Drowsy images classified as Non-Drowsy.

ResNet-18: 3,371 TP · 2,897 TN · 0 FN · 0 FP — zero misclassifications on this validation set.

Results Summary

Validation-set performance — CNN vs ResNet-18

Model	Accuracy	Precision	Recall	F1-Score
Custom CNN	99.97 %	100.00 %	99.94 %	99.97 %
ResNet-18	100.00 %	100.00 %	100.00 %	100.00 %

Note: Precision, Recall, and F1-Score for the CNN should be inserted from computed classification report output.

99.97 %

CNN Accuracy

100.00 %

ResNet-18 Accuracy

2

CNN misclassifications

0

ResNet-18 misclassifications

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Discussion & Limitations

Discussion — Interpreting Near-Perfect Accuracy

What high validation accuracy does and does not indicate

Both models achieved near-perfect accuracy on the DDD validation split. This result must be interpreted carefully.

Same-distribution evaluation

The validation set is a random split of DDD — drawn from the same recording session, same subject, same camera. High accuracy here does not confirm performance on new subjects, lighting conditions, or hardware.

Potential data leakage risk

If adjacent video frames from the same drowsiness episode appear in both training and validation, the model may learn subject-specific rather than generalisable patterns.

Transfer learning advantage confirmed

ResNet-18 converged in 8 epochs vs 12 for the CNN, and produced zero errors on the validation set — consistent with the established benefit of ImageNet pre-training for face-related tasks.

Practical deployment is a separate question

Real-world deployment requires head detection, varied lighting, motion blur handling, and testing on an independent population. These experiments do not address that.

Limitations

Constraints on validity and generalisation of findings

No independent test set

All reported metrics are from a validation split of DDD. There is no separate held-out test corpus or cross-dataset evaluation. True generalisation performance is unknown.

Single-subject dataset

DDD images originate from one individual. Demographic diversity (age, ethnicity, facial geometry) is not represented, limiting the generalisability of learned features.

Domain shift risk

Images are Viola–Jones face crops from a controlled environment. Deployment requires a real-time face detector and may encounter occlusions, extreme poses, and illumination changes not present in DDD.

No temporal context

Drowsiness is a progressive state that develops across time. A frame-level classifier cannot model head-nodding sequences, gradual eye closure, or PERCLOS — temporal methods would better capture these signals.

Mild class imbalance

The Drowsy class is ~15 % larger. No explicit resampling was applied; this could marginally affect recall for the minority class.

Training environment

GPU session time limits on Kaggle constrained hyperparameter search. A broader sweep might identify better configurations.

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Conclusion & Future Work

Conclusion

Summary of findings

Custom CNN

- Trained from scratch on DDD in 12 epochs.
- Achieved 99.97 % validation accuracy with 2 misclassifications.
- Demonstrates that a moderately deep CNN can learn effective drowsiness features without pre-training.

ResNet-18

- Fine-tuned from ImageNet weights in 8 epochs — faster convergence.
- Achieved 100.00 % validation accuracy with zero misclassifications.
- Confirms that transfer learning offers an efficiency and accuracy advantage for this task.

Both results are conditional on same-distribution validation. An independent test set and real-world deployment study are needed before deployment conclusions can be drawn.

Future Work

Directions for extending this research

1

External validation

Evaluate both models on an independent drowsiness dataset to measure generalisation beyond DDD. Suitable candidates include NTHU-DDD or the UTA Real-Life Drowsiness Dataset.

2

Temporal modelling

Replace the frame-level classifier with a sequence model (LSTM, 3D-CNN, or video transformer) that processes multiple consecutive frames to detect progressive drowsiness.

3

Eye and head landmarks

Incorporate PERCLOS (percentage of eyelid closure) or facial landmark tracking as complementary inputs to improve robustness under partial occlusion.

4

Lightweight architectures

Investigate MobileNet or EfficientNet variants for on-device inference. Evaluate the accuracy–efficiency trade-off explicitly before claiming real-time suitability.

5

Data diversity

Collect or augment DDD with images from multiple subjects, lighting conditions, and camera positions to reduce single-subject bias.

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Thank You!